Statistics Revision

Scatter Plots, Spread and Centre

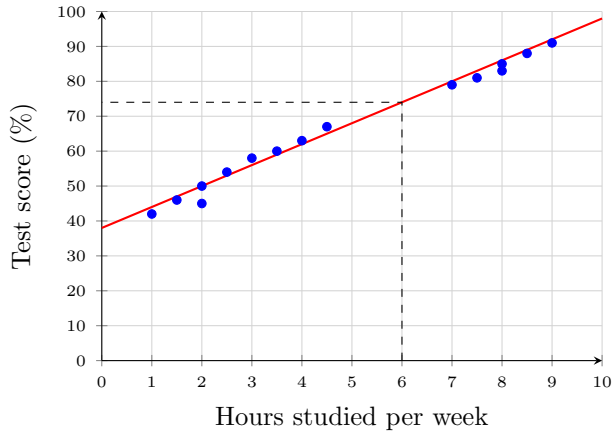
1 Scatter Plots and Lines of Best Fit

A **line of best fit** is a straight line drawn through scatter data that follows the overall trend, with roughly equal numbers of points above and below it. To **estimate** a value, you must draw this line first and then read your estimate *from the line* — not from whichever data point happens to be nearest.

Common mistake: Reading an estimate straight off the nearest data point, without drawing a line of best fit first. This is especially risky when there is a gap in the data near the value you need — without a line, there is nothing reliable to read from.

Worked example

The scatter graph shows hours studied per week against test score for 15 students. Estimate the test score for a student who studied for 6 hours.



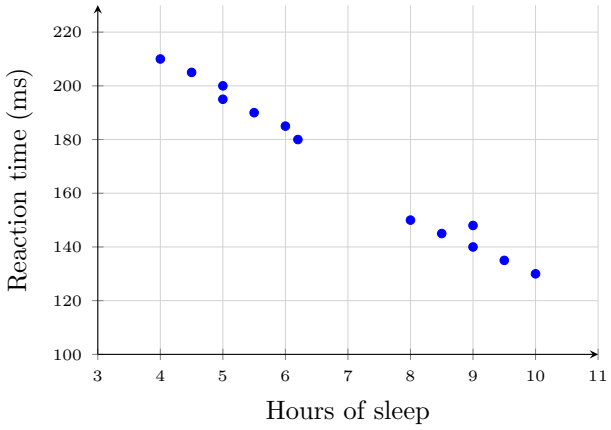
Step 1: Draw a line of best fit through the data (shown in red).
Step 2: Find 6 on the *hours studied* axis, go straight up to the line, then straight across to the *test score* axis (dashed lines).
Step 3: Read off the estimate: a student who studied for 6 hours is estimated to score about **74**%. (Notice there is no data point at 6 hours — the line is the only reliable way to estimate this.)

Try it yourself: Using the *same* line of best fit above, estimate the test score for a student who studied for 2.5 hours per week.

Practice

For each scatter graph: (a) draw a line of best fit by eye, (b) use your line to make the estimate asked for, and (c) answer the follow-up questions.

1. Hours of sleep vs reaction time

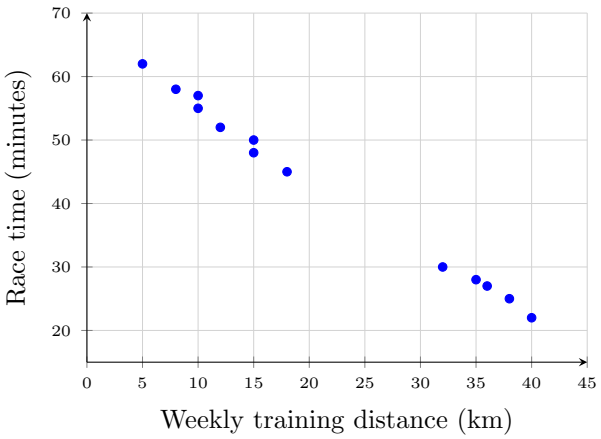


(a) Use your line of best fit to estimate the reaction time for someone who sleeps for 7 hours.

(b) Describe the correlation shown in the scatter graph (direction and strength).

(c) Why is a line of best fit especially necessary for part (a), rather than just reading off a nearby point?

2. Weekly training distance vs race time

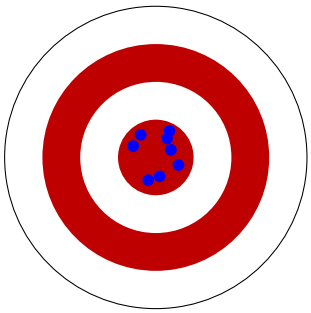


(a) Use your line of best fit to estimate the race time for a runner who trains 25 km per week.

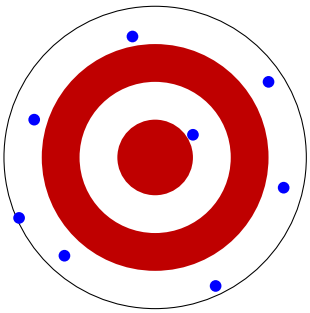
(b) Use your line to estimate the race time for a runner who trains 0 km per week. Explain why this particular estimate might *not* be reliable, even though you used a line of best fit.

2 Comparing Spread (Consistency)

Imagine two archers each shoot 8 arrows at a target.



Archer A: shots close together



Archer B: shots spread out

Archer A's arrows landed close together — **consistent** shooting, even though not every shot hit the bullseye. Archer B's arrows landed all over the target — **inconsistent** shooting: the more spread out the points are, the less consistent they are. Data works the same way.

Spread tells us how varied or consistent a data set is. The **range** (largest − smallest) and the **IQR** (interquartile range) both measure spread.

Smaller spread ⇒ values are closer together ⇒ *more consistent*.
Larger spread ⇒ values are more varied ⇒ *less consistent*.

Common mistake: Writing that the group with the *larger* range or IQR is "more consistent." It is the opposite: a larger range means the data values are more spread out and *less* alike, so that group is *less* consistent.

Worked example

Daily exercise times (minutes) for two groups over 8 days:

Group A: 25, 26, 27, 28, 29, 30, 31, 32 Group B: 10, 15, 20, 25, 40, 45, 50, 55

Which group has more consistent exercise times?

Range of Group A $= 32 - 25 = 7$ minutes. Range of Group B $= 55 - 10 = 45$ minutes.

Answer: Group A is more consistent, because it has the *smaller* range (7 minutes, compared to Group B's 45 minutes). This means Group A's exercise times are all similar to each other, while Group B's times vary a lot from day to day.

Practice

Judges' scores (out of 10) for two gymnasts over 10 routines:

Gymnast X: $7.2, 7.3, 7.4, 7.4, 7.5, 7.5, 7.6, 7.6, 7.7, 7.8$ Gymnast Y: $5.0, 5.5, 6.2, 6.8, 7.0, 8.5, 8.8, 9.0, 9.2, 9.5$

(a) Calculate the range for each gymnast.

(b) Which gymnast scored more consistently? Write a full answer, using your ranges, that explains *why* a smaller range means more consistent scoring.

3 Comparing Centre (Typical Value)

Measures of centre (mean, median) tell us about a *typical* value in a data set — this is a completely different question from spread/consistency.

Spread $\longrightarrow$ how consistent the data is. **Centre** $\longrightarrow$ what a typical value is.

Be careful: whether a *higher* or *lower* value is "better" depends on the context. A higher mean test score is better, but a lower mean race time is better.

Worked example 1 (higher is better)

Mean test scores: Class 1 = 74%, Class 2 = 61.5%. **Answer:** Class 1 typically scored higher, because its mean (74%) is greater than Class 2's mean (61.5%).

Worked example 2 (lower is better)

Mean race times: Swimmer P = 31.08 s, Swimmer Q = 29.53 s. **Answer:** Swimmer Q was typically faster, because a *lower* mean time means less time taken to finish the race — even though 29.53 is the smaller number, it is the better result here.

Mixed practice

Scores out of 30 for two quiz teams over 8 rounds:

Team Alpha: $15, 16, 16, 17, 17, 18, 18, 19$ Team Beta: $5, 8, 10, 12, 22, 25, 28, 30$

(a) Calculate the mean score for each team.

(b) Calculate the range for each team.

(c) Which team scored higher, on average? Justify your answer using your means.

(d) Which team was more consistent? Justify your answer using your ranges.

(e) Explain why your answers to (c) and (d) are both correct, even though they compare the teams in different ways.